Sentiment classification under processing constraints

Benchmark accuracy, temporal interpretation and local queueing

M.Sc. Data Science and AI

TweetEval and Bluesky experiments

Three questions guide the evaluation

01 How do the models compare on identical labeled posts?

02 How does model choice change the historical sentiment trend?

03 What happens to latency and completed posts as arrivals increase?

Two datasets serve different evaluation roles

12,284

Labeled Twitter test examples

45,615 training rows
2,000 validation rows

127,741

Cleaned Bluesky search posts

14 days of AI search results
194 requests reached the result cap

The comparison includes a simple baseline

01 VADER supplies the rule-based baseline.

02 TF-IDF with logistic regression tests a low-cost learned model.

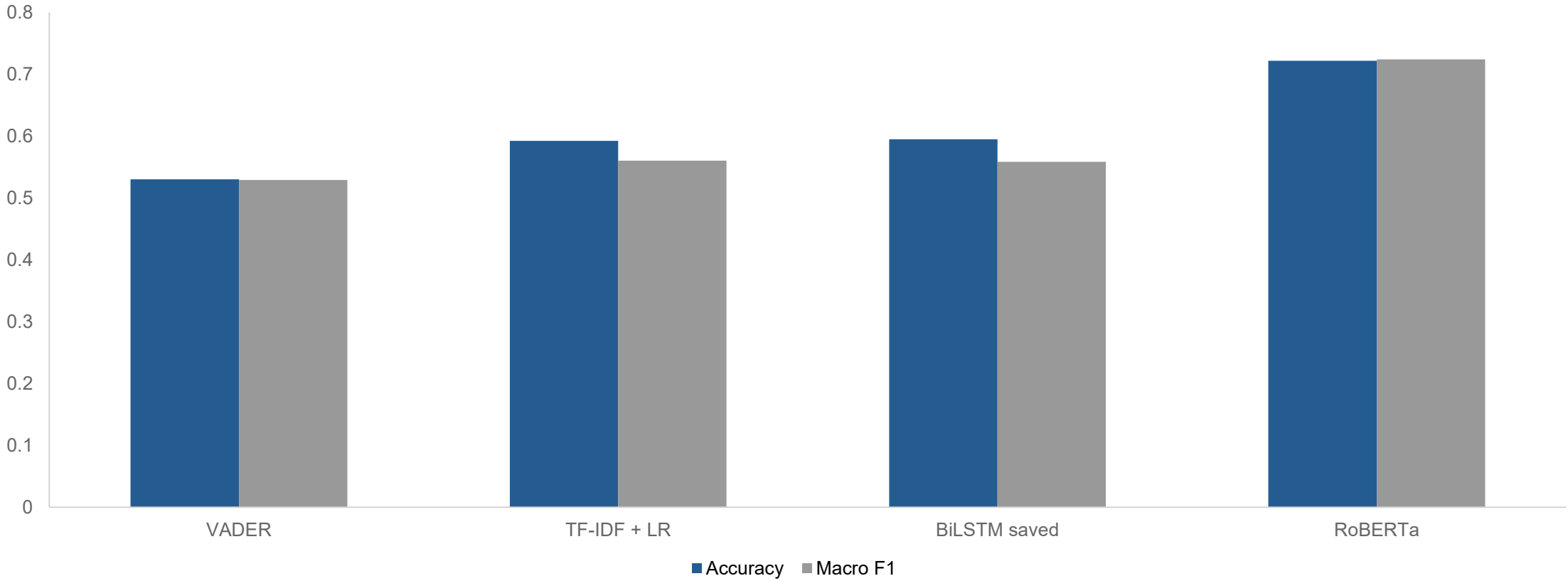
03 BiLSTM and Twitter RoBERTa supply the neural comparisons.

The implementation keeps evidence traceable

Collect and filter	Classify posts	Store in SQLite	Aggregate over time

Saved predictions, source hashes and timing records support reproduction.

RoBERTa leads the labeled benchmark



Score on a 0–1 scale. Neutral majority accuracy is 0.483.

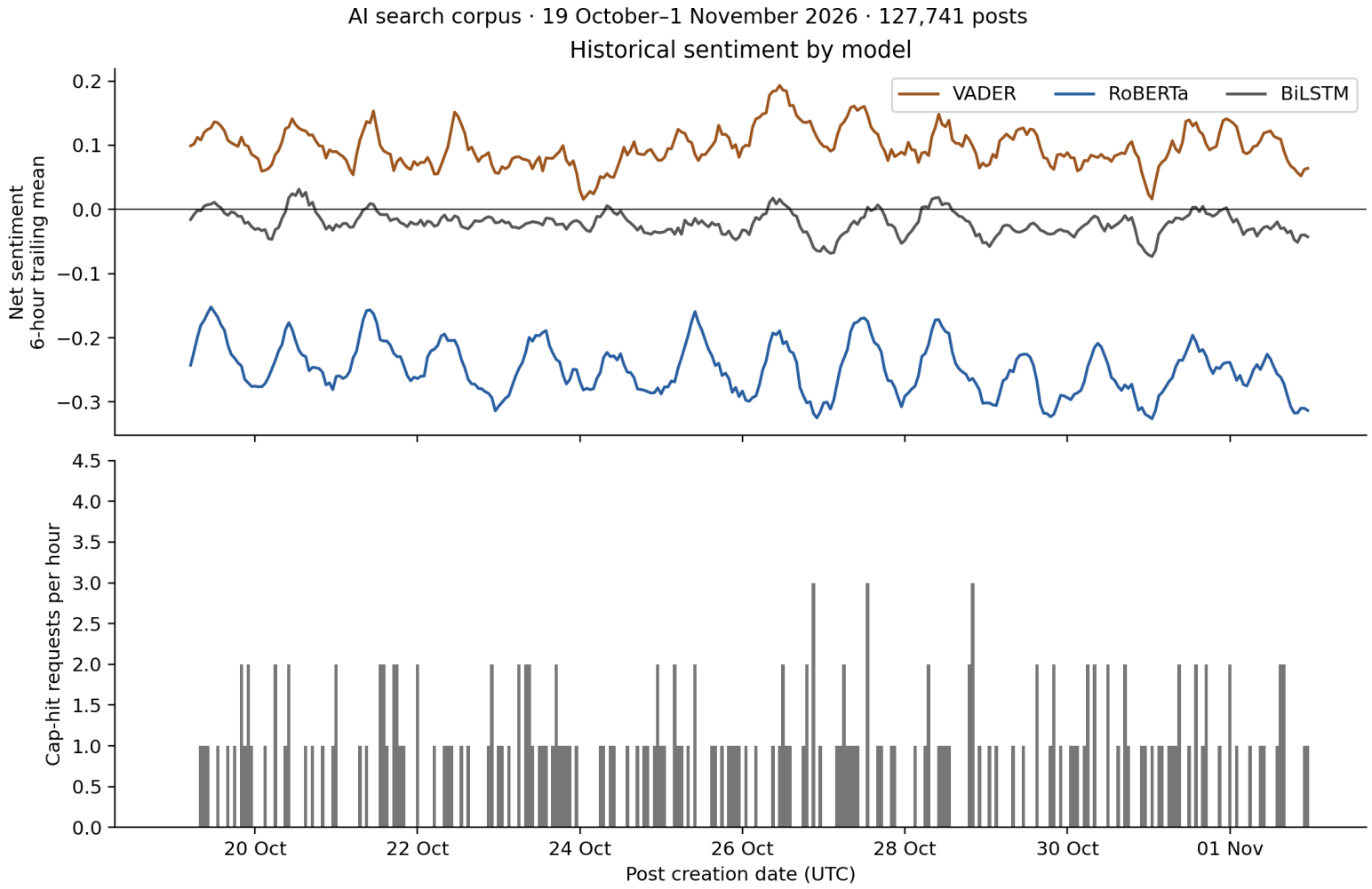
Small score differences need uncertainty

01 RoBERTa accuracy interval is 71.38% to 72.95%.

02 The TF-IDF versus BiLSTM accuracy-difference interval includes zero.

03 BiLSTM misses many negative examples by assigning neutral.

Classifier choice changes the temporal signal



Inference cost depends on batching and workload

26.45 ms

RoBERTa median run p50
for a single-post call

59.60 posts/s

RoBERTa median throughput
with batches of eight

These measurements describe warm local inference on the fixed test sample.

Controlled replay exposes queue overflow

Model	Offered posts/s	Stored of 288	Dropped of 288
VADER	10	288	0
VADER	40	288	0
VADER	80	288	0
RoBERTa	10	191	97
RoBERTa	40	79	209
RoBERTa	80	72	216

Latency of completed posts must be interpreted alongside dropped arrivals.

The findings have explicit boundaries

01 Bluesky labels are unverified, and search coverage is incomplete.

02 Pretraining histories differ, and bootstrap assumes independent rows.

03 Short local replay omits live delivery and long-running system effects.

The study connects model quality with monitoring behaviour

01 RoBERTa has the highest observed labeled benchmark scores.

02 The same historical posts produce different sentiment directions.

03 Processing constraints affect which arriving posts reach storage.